Yusef Syed

Software Engineer: AI agent security & evals | 14 merged PRs: Microsoft, Apache Arrow, W&B
yusef.syed@mail.utoronto.ca | github.com/YusefSyed | linkedin.com/in/yusef-syed-4b811a289
Toronto, ON | U.S. & Canadian citizen, no sponsorship needed | Will relocate to San Francisco or New York
Available full-time from January 2027: security engineer, evals engineer, AI red team, AI safety research engineer

Open Source

14 merged PRs (AI-assisted): Microsoft, Apache Arrow, W&B, Inspect Scout, Kornia | Full list **Sep – Oct 2026**

- Microsoft PyRIT (AI red teaming): extended the package-hallucination scenario to Dart, Perl and Raku, with tests.
- W&B RAI Toolkit: landed five evaluation fixes, including keeping a failed prediction's error text from scoring as a pass.
- Apache Arrow: extended four PyArrow slice APIs to accept Arrow integer scalars, with regression tests.

Publications

ToolEffectBench: Scoring Prompt Injection on Tool-Using Agents from Persisted State **2026**

Sole author | Zenodo v0.7.1, doi:10.5281/zenodo.23966727 | code & logs

- Built an Inspect benchmark scoring prompt injection from a sandbox's database state, not transcripts (188 samples, 17 models, 5,628 scored runs); on one synthetic task, agents' final reports denied at least **13 of 232** recorded forbidden publications.

Checks for Checkers: Sealed Paired Comparisons of AI Agent Verification Tools **2026**

Sole author | Zenodo v3, doi:10.5281/zenodo.23966747

- Sealed, blind comparison of my startup's tool and an open-source framework: no lead (83 vs 88 of 96, p = 0.30).

Projects

effect-tracer | *Rust, Linux ptrace/seccomp, Ed25519* | github.com/YusefSyed/effect-tracer **2026**

- Built a tracer that signs a process tree's file writes, connects, DNS and execs into a hash chain: **19 of 20** seeded bypasses caught at 1.1–1.9x overhead in seccomp mode (20 of 20 at 5–298x in full mode); 0 undeclared events in 30 clean runs.

Experience

Scopehaven | github.com/YusefSyed/scopehaven-oss **Oct 2026 – Present**

Software Engineer (creator, solo project); security testing for AI agents

- Built **Scopehaven**, a Node.js CLI and GitHub Action that tests whether an AI agent touched the wrong account, tenant or refund, decided from the app's own records rather than the agent's report.
- Released the **Apache-2.0 open edition** (v0.7.0: 4,800+ tests, zero dependencies, GitHub Action, MCP test target); on my synthetic test app the vulnerable build fails 26 of 44 checks.
- Designed signed, tamper-evident evidence (DSSE/in-toto exports, a hash-chained ledger) and a stdlib-Python checker that re-derives verdicts from kept records: 88 of 88 on both demo builds (synthetic app, no model), 8 of 8 tampered copies rejected.
- I build with AI coding agents that I direct and review; on Scopehaven I ran the merge gate as the only human (exact-head CI, a second AI review, secret scanning): 57 verdicts asked for changes.

Independent Security Research **Oct 2026 – Present**

Security Researcher *Remote*

- Filed 15 private security advisories and 2 HackerOne reports (authorization, credential and data-exposure flaws) with 10 open-source projects, each reproduced 3 of 3 with controls first by an automated harness; all awaiting maintainer triage.

Education

University of Toronto **Expected May 2030**

B.Sc., Math & Physical Sciences (first year); intended Math + CS; would leave for the right role

Coursework: Linear Algebra, Proofs & Calculus, Statistics; Data Structures (LCCC, dual enrollment)

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Swift
AI & security: application security (AppSec), AI security, AI red teaming, prompt-injection evals, vulnerability reproduction, access-control testing (IDOR, RLS), Inspect, PyRIT, OWASP LLM/Agentic Top 10, MITRE ATLAS
Systems & tools: Node.js, React Native, SwiftUI, PostgreSQL, Docker, GitHub Actions, PyTorch, Claude Code

Other Projects & Activities

Providence | *Swift* | **1st Place, Aramco Americas Challenge** (Best Beginner Hack), Hack the North **Sep 2026**

Aesthetics AI & **Tiraz** | *TypeScript, React Native* | two iOS apps on the **App Store** | Study **2026**

USA Powerlifting (60 kg, 2025): #2 in PA, #19 nationally | **Feeding the Valley**, VP: $20,000 for food banks